

Robert C. Bridges

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PROFESSIONAL SUMMARY

Mechanical and aerospace engineer with hands-on CAD design experience (Fusion 360), GD&T, and machine shop fabrication, plus experience supporting engineering and quality teams through bill of materials (BoM) analysis, issue documentation, and troubleshooting. Project work has taken designs from concept through fabrication and test with complete technical documentation. Interprets engineering drawings, works with cross-functional teams, and teaches others to use tools. Proficient in MS Office, Python, and MATLAB.

EDUCATION

Cornell University, College of Engineering, Ithaca, NY

Master of Engineering, Aerospace Engineering

May 2026

Bachelor of Science, Mechanical Engineering

Dec 2025

Relevant Coursework: Mechanical Metallurgy; Space Structures; Spacecraft Technology and Systems; Introduction to Spaceflight Mechanics; Propulsion, Aircraft and Rockets; Combustion Processes; Plasmas for Propulsion

TECHNICAL SKILLS

- **CAD & Design:** Fusion 360 (parametric modeling; applies to Creo, SolidWorks, AutoCAD), Ansys (FEA/CFD), GD&T, engineering drawing interpretation, machine and tool design
- **Machine Tools & Fabrication:** hand tools and machine tools (Red Apron Machine Shop), Micro Vu inspection, 3D printing (PLA, TPU, resin), prototype assembly
- **Materials & Structures:** engineering materials, mechanical metallurgy, composite materials (honeycomb sandwich panels, carbon fiber layups), structural analysis
- **Documentation & Software:** BoM, technical documentation, design reviews, MS Office Suite, Adobe Creative Suite, Python, MATLAB, Java

PROFESSIONAL EXPERIENCE

Abbott Point of Care, Princeton, NJ

Project Engineering Intern

May - Aug 2022 & 2023

- Built and maintained a database of custom parts used across Abbott's medical devices, streamlining search and documentation for engineering teams (2023).
- Analyzed engineering bills of materials (BoM) and product structures for hundreds of components to support design and manufacturing decisions.
- Used precision measurement equipment, including Micro Vu, to inspect circuit boards; updated and calibrated TEAM, a company-wide database for tracking product errors (2022).
- Documented product issues and test results to strengthen troubleshooting and quality processes throughout the design cycle.

Princeton University, Facilities Department, Princeton, NJ

Engineering Intern

May - Aug 2025 & 2026

- Coordinated communication between project managers and on-site staff, producing daily progress reports to keep stakeholders aligned.
- Read and interpreted technical architectural plans and drafted a granite sign engraving template to support facility construction projects.

ENGINEERING PROJECTS

Robotic Bat Research, Master's Project

Aug 2025 - May 2026

- Designing and building a robotic bat, including its structure, circuitry, motors, and wing membrane, carrying the hardware from concept through fabrication and test.
- Fabricate and iterate on structural components using PLA, TPU, and resin 3D printing.

Focus Glasses

Aug 2025 - Dec 2025

- Led market research and delivered a Product Design Specification with a 4-person team for glasses that keep users focused using invisible spectral flicker; completed finite element analysis to confirm safety.
- Voted best new product in the class after presenting at a public exhibit.

Original Spacecraft Mission Design, F.O.C.A

Jan - May 2026

- Designed a spacecraft to intercept foreign objects in orbit, using prior flight missions as a baseline for the vehicle architecture.
- Authored SRR, SDR, and PDR documentation to validate the design and defined a full mission life timeline.

Additional Projects

- **Spin Wheel, Senior Design** (Jan - May 2025): designed, prototyped, and filed a patent application for a novel computer scroll wheel.
- **Chainsaw Dissection** (Aug - Dec 2024): deconstructed a chainsaw to analyze its mechanical components and fluid dynamic properties.
- **Van Allen Belt Electron Simulation** (Jan - May 2024): simulated electron acceleration in the outer Van Allen Belt during a May 2024 solar storm (up to 2.4 MeV).

TEACHING & MENTORSHIP

- Teaching Assistant, Introduction to Spaceflight Mechanics & Introduction to Mechanical Design, Cornell MAE Department: graded assignments, led weekly labs, hosted office hours, and taught students aerospace and mechanical design (2025 - Present).
- Grader, Thermodynamics, Cornell MAE Department: created rubrics, graded homework, and reported to faculty (2024).

CAMPUS INVOLVEMENT

Ultimate Frisbee, Social Media Chair & Captain (2022-26) | Orientation Leader (2022) | Mediocre Melodies (2023-24) | MEng Content Creator (2026-Present)